Minimised Evidence Sharing for Cross-Bank Money-Laundering Detection: Pre-Registered Sealed Tests and Fresh Replication on AMLworld

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***Abstract*—Abstract to be written (step 4). Headline numbers: HI-Large linked scheme detection 0.063 → 0.534 with minimised evidence sharing (0.92 of a pooled relational oracle's gain), replicated on two never-opened test windows; 8.5× fewer alerts per linked scheme; a gated alert flag closes the one leak the audit found, at no detection cost.**

***Index Terms*—anti-money laundering, privacy-enhancing technologies, inter-bank information sharing, pre-registration, oblivious pseudorandom functions**

I. Introduction

To be written.

II. Related Work

See `related_work.md` for the draft text and Table I.

III. System

To be written: banks, tokens (OPRF [10]), evidence records, sharing levels S0–S4, progressive disclosure, gated flags (randomised response [11]).

IV. Evaluation Method

To be written: AMLworld [9], federation, pre-registration [12], sealed tests, Holm correction [13], amendments. Table II.

V. Results

A. Detection

Fig. 1, Table III.

B. Robustness

Fig. 2, Table IV.

C. Privacy

Fig. 3.

D. Case Quality and Analyst Workload

Fig. 4, Table V.

E. Case Narratives

Table VI.

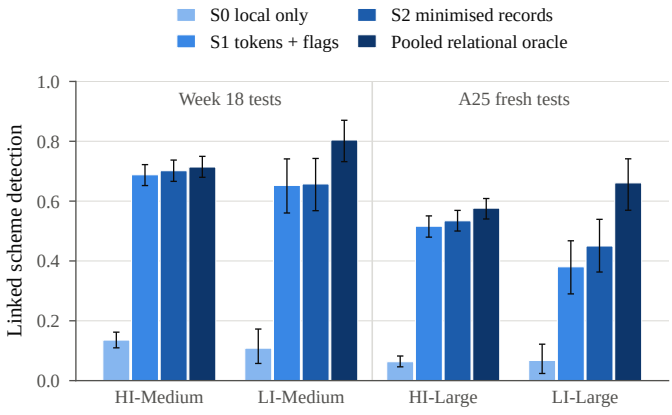


Fig. 1. Linked scheme detection on the four sealed tests by sharing level: local only (S0), tokens and alert flags (S1), minimised evidence records (S2), and a pooled relational oracle that sees every bank's data. Bars: mean over seeds; whiskers: 95% scheme bootstrap intervals. HI- and LI-Medium were decided on the week-18 tests; HI- and LI-Large are the A25 windows, never opened during development.

VI. Limitations

To be written: H3a, PT2, small LI windows, synthetic data, automated narrative judge.

VII. Conclusion

To be written.

References

[1] T. Suzumura, Y. Zhou, N. Baracaldo, G. Ye, K. Houck, R. Kawahara, A. Anwar, L. L. Stavarache, Y. Watanabe, P. Loyola, D. Klyashtorny, H. Ludwig, and K. Bhaskaran, "Towards federated graph learning for collaborative financial crimes detection," NeurIPS 2019 Workshop on Robust AI in Financial Services; arXiv:1909.12946, 2019.

[2] S. Arora, A. Beams, P. Chatzigiannis, S. Meiser, K. Patel, S. Raghuraman, P. Rindal, H. Shah, Y. Wang, Y. Wu, H. Yang, and M. Zamani, "Privacy-preserving financial

TABLE I
COLLABORATIVE ANTI-MONEY-LAUNDERING AND FINANCIAL-CRIME DETECTION: WHAT IS SHARED, HOW IT IS PROTECTED, AND HOW IT IS EVALUATED

Work	Shared between institutions	Protection	Data	Evaluation	Privacy evidence
Suzumura et al. [1]	Model parameters (global graph features planned)	Federated learning; DP/MPC proposed, not evaluated	Simulated, six banks (FCA TechSprint)	One test split, prototype	None
Arora et al. [2]	Bank account flags, inside a payment hub's training	Federated learning + MPC (oblivious transfer, secret sharing), noise	PETs Prize synthetic payments	Challenge test environment	Design argument
Abadi et al. [3]	Identity alignment and model updates	Private set intersection + secure federated learning	Synthetic payments	Scalability and accuracy, prototype	Security proof
Khan et al. [4]	Embeddings and gradients	Differential privacy + secure summation	PETs Prize synthetic, AMLSim	One test split; best test epoch reported	Formal DP bound
van Egmond et al. [5]	Encrypted risk scores, propagated across banks	Secure multi-party computation	One bank's transactions	Precision at fixed recall	Design argument
Brand et al. [6]	Encrypted flow-tracing indicators	Homomorphic encryption	Nation-scale deployment	Run time only	Design argument
BIS Innovation Hub [7]	Collaborative analysis and learning	Privacy-enhancing technologies	Synthetic, national and cross-border	Scenario comparison	Not reported
Zheng et al. [8]	Model updates and aggregated node embeddings	Federated learning (no formal guarantee)	AMLworld HI-Small subset [9]	One split (5% train)	None
This work	Minimised evidence records keyed by OPRF tokens; alert flags, gated	Pseudonymous tokens (OPRF), data minimisation, flag gating; LDP and progressive disclosure as options	AMLworld HI/LI Medium and two never-opened Large windows [9]	Pre-registered sealed tests, fresh replication, 95% CIs, Holm	Empirical attacks (4 targets) and token re-identification

TABLE II
TEST SETS

Test set	Transactions (millions)	Laundering rows	Schemes	Cross-member test schemes
HI-Medium	31.9	31,146	2,756	480
LI-Medium	31.3	15,270	456	91
HI-Large	30.3	39,798	6,638	559
LI-Large	29.7	18,037	896	78

Each set is a 16-day dense window of AMLworld, split by 6-h window 60/20/20 into train, validation and test with one-window gaps; eight institutions (federation B8) hold every account. Cross-member test schemes: laundering schemes in the test partition that touch two or more institutions. Medium: the week-18 sealed tests; Large: the A25 fresh tests.

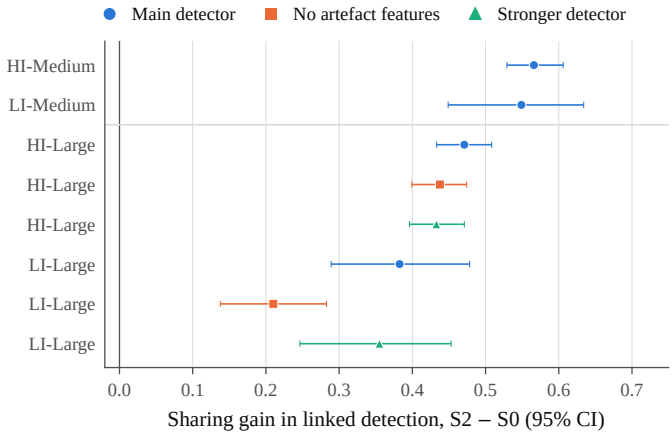


Fig. 2. Sharing gain (S2 − S0 linked detection) with 95% intervals. Main detector on all four tests; on the Large tests also without the suspected artefact features (payment-format shares, self-transfers) and with a stronger transaction-level detector that outperforms both published gradient-boosting baselines on the AMLworld benchmark (Table IV).

anomaly detection via federated learning & multi-party computation," arXiv:2310.04546, 2023.

[3] A. Abadi, B. Doyle, F. Gini, K. Guinamard, S. K. Murakonda, J. Liddell, P. Mellor, S. J. Murdoch, M. Naseri, H. Page, G. Theodorakopoulos, and S. Weller, "Starlit: Privacy-preserving federated learning to enhance financial fraud detection," Cryptology ePrint Archive, Paper 2024/090, 2024.

[4] M. S. I. Khan, A. Gupta, O. Seneviratne, and S. Patterson, "Fed-RD: Privacy-preserving federated learning for financial crime detection," arXiv:2408.01609, 2024.

[5] M. B. van Egmond, V. Dunning, S. van den Berg, T. Rooijakkers, A. Sangers, T. Poppe, and J. Veldsink, "Privacy-preserving anti-money laundering using secure multi-party computation," in *Financial Cryptography and Data Security (FC 2024)*, 2024, cryptology ePrint Archive, Paper 2024/065.

[6] M. Brand, H. Ivey-Law, and T. Churchill, "FinTracer: A privacy-preserving mechanism for tracing electronic money," Cryptology ePrint Archive, Paper 2023/649, 2023.

[7] BIS Innovation Hub, "Project Aurora: the power of data, technology and collaboration to combat money laundering across institutions and borders," Bank for

TABLE III
PRE-REGISTERED CLAIMS ON THE SEALED TESTS

Pre-registered claim	HI-Medium	LI-Medium	HI-Large	LI-Large
H1a: S2 − S0 linked detection	0.57 [0.53, 0.61] ✓	0.55 [0.45, 0.63] ✓	0.47 [0.43, 0.51] ✓	0.38 [0.29, 0.48] ✓
H1b: share of oracle gain ($\geq$0.60)	0.98 [0.94, 1.03] ✓	0.79 [0.69, 0.88] ✓	0.92 [0.87, 0.97] ✓	0.64 [0.52, 0.77] ✓
H1c: plain flag, fan-in advantage	−0.002 [−0.008, 0.003] ✓	0.092 [0.079, 0.103] ✗	0.001 [−0.002, 0.003] ✓	0.043 [0.031, 0.059] ✗
gated flag (A25)	—	—	0.001 [−0.001, 0.003] ✓	0.004 [0.000, 0.007] ✓
H1d: PD gain share, least disclosure	0.98 at 0.07 ✓	0.99 at 0.01 ✓	0.95 at 0.03 ✓	0.81 at $<$0.01 ✓
H3a: attribution above baselines	✓	✗	✗	✗
H3b: narrative grounding ($\geq$0.95)	0.86 [0.81, 0.90] ✗	0.92 [0.84, 0.96] ✗	1.00 [0.98, 1.00] ✓	0.95 [0.87, 0.98] ✓
PT2: hashing ÷ OPRF re-id. ($\geq$10)	19.9 ✓	8.7 ✗	8.1 ✗	9.3 ✗

Sealed tests; 95% intervals from a scheme bootstrap. H1c: upper 95% bound of (attack − prior) TPR at 1% FPR at most 0.05 on every attack target (A2, A3); the value shown is the binding target, A3 fan-in, and the mark is the verdict over all targets and, for the plain flag, the S3 levels with $\varepsilon \leq 2$. H1d: PD configuration reaching $\geq$0.80 of S2's gain with the smallest share of S2's disclosed accounts. H3b on Medium: free-text narratives (A23); on Large: select-then-template (N4). Medium columns: week-18 tests; Large: A25 fresh tests.

TABLE IV
DETECTOR AGAINST PUBLISHED RESULTS

Model	HI-Small	LI-Small	HI-Medium	LI-Medium
Ours, account-window detector	48.20 ± 0.69	23.21 ± 2.80	42.67 ± 2.76	24.08 ± 2.01
Ours, transaction model	64.25 ± 1.24	**34.99 ± 1.51**	**67.01 ± 2.27**	34.34 ± 1.55
LightGBM + graph features [14]	62.86	20.83	59.48	20.85
XGBoost + graph features [14]	63.23	27.30	65.70	28.16
Multi-PNA + EU [14]	**68.16**	33.07	66.48	**36.07**

Minority-class F1 (%) on AMLworld's published split (all transactions, by day 60/20/20), threshold chosen on validation; ours: mean ± s.d. over 5 seeds; best per column in bold. Pooled data in every row (the published models see all banks).

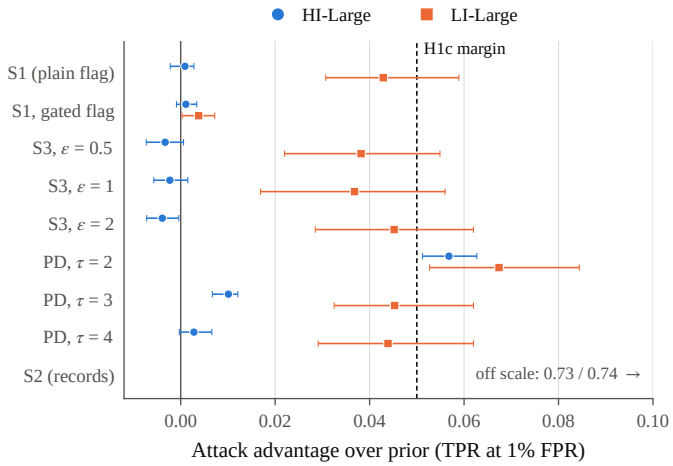


Fig. 3. Leakage audit on the Large tests: advantage of a strong attacker over a prior-only attacker in inferring whether a queried account has top-decile fan-in (the binding target), true-positive rate at 1% false-positive rate, with 95% intervals. The dashed line is the pre-registered margin (upper bound $\leq$ 0.05). On LI-Large the plain alert flag and every level that carries it exceed the margin; the gated flag chosen on validation does not.

TABLE V
CASE QUALITY AND ANALYST WORKLOAD

Measure	HI-Large		LI-Large	
	S0	S2	S0	S2
Scheme accounts in the case	0.47	0.77	0.48	0.73
Cases per linked scheme	1.51	1.05	1.49	1.06
Alerts per linked scheme	153	18	944	140

Per detected scheme, seed-averaged: share of the scheme's accounts in its largest case (completeness), number of cases it is split into (fragmentation), and alerts raised per scheme linked (analyst workload). A25 fresh tests, seeds 0–4.

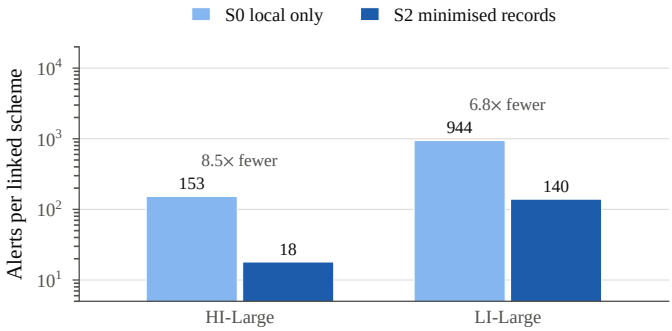


Fig. 4. Alerts an analyst reviews per laundering scheme linked, local-only monitoring (S0) versus minimised evidence sharing (S2), on the Large tests (logarithmic axis).

TABLE VI
CASE NARRATIVES: LOCAL MODELS AND SELECT-THEN-TEMPLATE

Design	Grounding	Strict	Faithful
HI-Medium validation, 200 cases (judge: 30)			
N0: `llama3.2:3b`, free text	0.96	0.13	21/30
N1: `qwen2.5:7b`, free text	0.97	0.10	26/30
N2: `llama3.1:8b`, free text	0.99	0.06	29/30
N3: `qwen3:8b`, free text	0.98	0.22	23/30
N4: select-then-template (N3)	0.99	0.83	30/30
N4 on the A25 fresh tests			
HI-Large	1.00	0.82	—
LI-Large	0.95	0.56	—

Grounding (H3b): first reply is valid, keeps the attributed typology and cites only evidence fields. Strict: also 1–4 fields, every number from the evidence, claim consistency and scope. Faithful: automated judge (one rater, not blind) on the 30-case sheet; delivered text after retry and template fallback.

International Settlements, Tech. Rep., May 2023.

[8] L. Zheng, J. Ni, C. Zobel, and J. R. Birge, "Networked markets, fragmented data: Adaptive graph learn-

ing for customer risk analytics and policy design," arXiv:2512.24487, 2025.

[9] E. Altman, J. Blanuša, L. von Niederäusern, B. Egressy, A. Anghel, and K. Atasu, "Realistic synthetic financial transactions for anti-money laundering models," in *Advances in Neural Information Processing Systems 36 (NeurIPS 2023), Datasets and Benchmarks Track*, 2023.

[10] A. Davidson, A. Faz-Hernández, N. Sullivan, and C. A. Wood, "Oblivious pseudorandom functions (OPRFs) using prime-order groups," RFC 9497, IETF, Dec. 2023.

[11] S. L. Warner, "Randomized response: A survey technique for eliminating evasive answer bias," *Journal of the American Statistical Association*, vol. 60, no. 309, pp. 63–69, 1965.

[12] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, and D. T. Mellor, "The preregistration revolution," *Proceedings of the National Academy of Sciences*, vol. 115, no. 11, pp. 2600–2606, 2018.

[13] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[14] B. Egressy, L. von Niederäusern, J. Blanuša, E. Altman, R. Wattenhofer, and K. Atasu, "Provably powerful graph neural networks for directed multigraphs," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, 2024.